

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks available |     |     |       |       |      |
|----------|-----|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 8.       | (a) |  |  | <p>transfer of electrons / correct electronic structure of ions (1)<br/>charges on ions (1)</p> <div><div><div>Al <math>\times</math><br/><math>\times</math></div><div>Al <math>\times</math><br/><math>\times</math></div></div><div><div><math>\ddot{\text{O}}:</math></div><div><math>\ddot{\text{O}}:</math></div><div><math>\ddot{\text{O}}:</math></div></div><div><math>\longrightarrow</math></div><div><div><math>[\text{Al}]^{3+}</math></div><div><math>[\text{Al}]^{3+}</math></div></div><div><div><math>[\text{x}\ddot{\text{O}}:]^{2-}</math></div><div><math>[\text{x}\ddot{\text{O}}:]^{2-}</math></div><div><math>[\text{x}\ddot{\text{O}}:]^{2-}</math></div></div></div> |                 | 2   |     | 2     |       |      |
|          | (b) |  |  | <p><math>n = \frac{pV}{RT}</math> (1)</p> <p>award (1) for correct units for all of volume, temperature and pressure</p> <p><math>V = 6.08 \times 10^{-5} \text{ m}^3</math><br/><math>T = 493 \text{ K}</math><br/><math>p = 1.01 \times 10^5 \text{ Pa}</math></p> <p><math>n = \frac{1.01 \times 10^5 \times 6.08 \times 10^{-5}}{8.31 \times 493}</math></p> <p><math>n = 1.499 \times 10^{-3} \text{ mol}</math> (1)</p> <p><math>M_r = \frac{0.400}{0.001499} = 266.8 \quad \Rightarrow M_r \text{ of Al}_2\text{Cl}_6 \text{ is } 267</math> (1)</p>                                                                                                                                   | 1               |     |     |       |       |      |
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 | 2   | 1   | 4     | 3     |      |